Sherwin-Williams Recommendation System: A Data-Driven Approach to Paint Selection

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1 Introduction

My project seeks to utilize publicly available product information to support customers in selecting appropriate Sherwin-Williams paint products based on project requirements and product constraints, streamlining the decision-making process.

For decades within the paint industry, lead paint served as the industry standard because it provided a cheap, easy, and convenient solution for homeowners. However, over time, it became apparent that this convenience came at a high cost as lead exposure became linked to serious health problems, including neurological damage, reproductive issues, muscle and joint pain, etc. Thus, in 1978, the U.S. addressed these concerns by implementing the Consumer Product Safety Commission ban, which prohibited the sale of lead-based paint for residential use. After the ban, the paint industry shifted toward safer formulations and expanded into broader and newer categories such as low-VOC products, antimicrobial coatings, stain-blocking technologies, and other specialized options. This diversification of products created a much broader paint scene. As a result, modern homeowners now face an overwhelming number of paint choices. Even with product data sheets readily available, interpreting this information and making confident, informed decisions has become a new challenge within the industry. .

2 Resources and Methods

As previously mentioned, my project aims to utilize publicly available product information, as Sherwin-Williams provides three types of documentation for each paint product.

Safety Data Sheet: covers safety and hazard information required by OSHA

Product Data Sheet: provides technical and application details for proper product use

Environmental Data Sheet: reports environmental and regulatory data such as VOC content.

My project will mainly aim to utilize the PDS in conjunction with the Paint Grades Guide. These manufacturer sheets containing structured information such as durability, coverage, VOC levels, drying time, and mold/mildew resistance. Aiming to reduce the amount of data cleaning, I seek to create my own database within Google Sheets and later import this into Python for further analysis of the categories.

Timeline:

Checkpoint Date	Task
2/12/26	Data loading and cleaning
2/26/26	Full Database Creation
3/19/26	Apply K-Means Clustering Techniques and begin Flask development
4/16/26	Aim to enter final development, testing, and refinement phase of project

My research plan is practical in the sense that the data sources have already been identified. Additionally, the scope of the project is intentionally limited to homeowner-grade paints. By excluding industrial and automotive-grade paints, the project remains focused and feasible within the given constraints.

3 Works Cited

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